

Applications of Radian Units for the Measure of Angles (Homework)

① Find the length of an arc on a circle with radius 12 centimeters and central angle $3\pi/8$.

② Find the length of an arc on a circle with radius 8.6 feet and central angle $7\pi/9$. Round to the nearest hundredth.

③ Find the area of a sector of a circle with radius 62 millimeters and central angle $3\pi/7$. Round to the nearest tenth.

④ Find the area of a sector of a circle with radius 5 miles and central angle $18\pi/5$. Round to the nearest whole number.

⑤ An object moves on a circular path with an angular velocity of 3π rad/min. What is the object's linear velocity if the radius of the path is 7.7 inches? Round to the nearest thousandth.

⑥ An object moves on a circular path with a linear velocity of 52 m/s . What is the object's angular velocity if the radius of the path is 94 meters? Round to the nearest thousandth.

⑦ A helicopter blade rotates at 300 rpm. Convert this rate to radians per second.

⑧ A Frisbee rotates at 15π radians per second. Convert this rate to rpm.

⑨ The hour hand on a clock rotates $360^\circ/\text{h}$. Convert this rate to rpm.

⑩ A disc rotates at 135 rpm. If the diameter of the disc is 18 centimeters, how fast does the edge farthest from the center move? Round to the nearest whole number.

⑪ An alternator converts rotational energy to electrical energy by spinning wires that are wound around a circular frame. The alternator completes one rotation in 0.043 seconds. If the radius of the device is 3 inches, what is the linear velocity of the outer edge? Round to the nearest hundredth.



⑫ A Ferris wheel makes four complete rotations in 48 seconds. If a woman is standing 17 feet from the center of the wheel, what is her linear velocity in feet per minute? Round to the nearest whole number.

⑬ Holes are drilled into brake discs to dissipate heat. One hole is 13 centimeters from the center of the disc and a different hole is 8 centimeters from the center. If the hole farther away from the center rotates 35 centimeters per second, what is the angular velocity of the disc and how fast does the second hole move? Round to the nearest hundredth.

14) A band performs on a rotating stage. The guitarist rotates 7 millimeters per second and he is 800 millimeters from the axis of rotation. The drummer is 200 millimeters from the axis of rotation. What is the angular velocity of the stage and what is the linear velocity of the drummer?

Homework Answers

① $\frac{9\pi}{2} \text{ cm}$

⑫ 534 ft/min

② 21.01 ft

⑬ 2.69 rad/s and 21.54 cm/s

③ $2,587.8 \text{ mm}^2$

⑭ 0.00875 rad/s and 1.75 mm/s

④ 141 mi^2

⑤ 72.571 in/min

⑥ 0.553 rad/s

⑦ $10\pi \text{ rad/s}$

⑧ 450 rpm

⑨ $\frac{1}{60} \text{ rpm}$ or $0.01\bar{6} \text{ rpm}$

⑩ $7,634 \text{ cm/min}$

⑪ 438.36 in/s